



ARCHITECTURE WHITEPAPER

BITGET AI BASE CAMP HACKATHON — SEASON 2

LUNARIS TERMINAL

*Beyond “AI Slop”: Designing an Adversarial, Multi-Agent Autonomous
Trading System with Deterministic Risk Hard-Guards and Merkle-Verified Auditability*

PROJECT

LunarIS Terminal — AI x Tokenized US Stock & Crypto Trading Desk

Live app: <https://lunaristerminal.vercel.app/>

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Executive Abstract

The generative AI boom has flooded decentralized finance and algorithmic trading with superficial wrappers — often termed “AI slop.” These applications typically connect a raw Large Language Model (LLM) to a single prompt, ask it for price predictions, and execute trades without structural risk boundaries, adversarial critique, or cryptographic auditability. When markets suffer regime shifts or liquidity shocks, these unconstrained systems fail catastrophically due to sycophancy, stochastic drift, and unmitigated hallucinations.

LUNARIS Terminal is an architectural antidote to this paradigm. Engineered for cross-asset deployment across spot crypto (Bitget) and mega-cap tech equities, LUNARIS replaces the vulnerable single-model prompt with an asynchronous, multi-agent cognitive architecture, governed by deterministic execution gates that no generative model can override.

This whitepaper details the architectural philosophy, technical implementation, and security models that separate LUNARIS from conventional AI financial prototypes.

1. The Problem Space: Why Generative AI Fails in Financial Markets

1.1 The Pitfalls of Single-Agent LLM Trading (“AI Slop”)

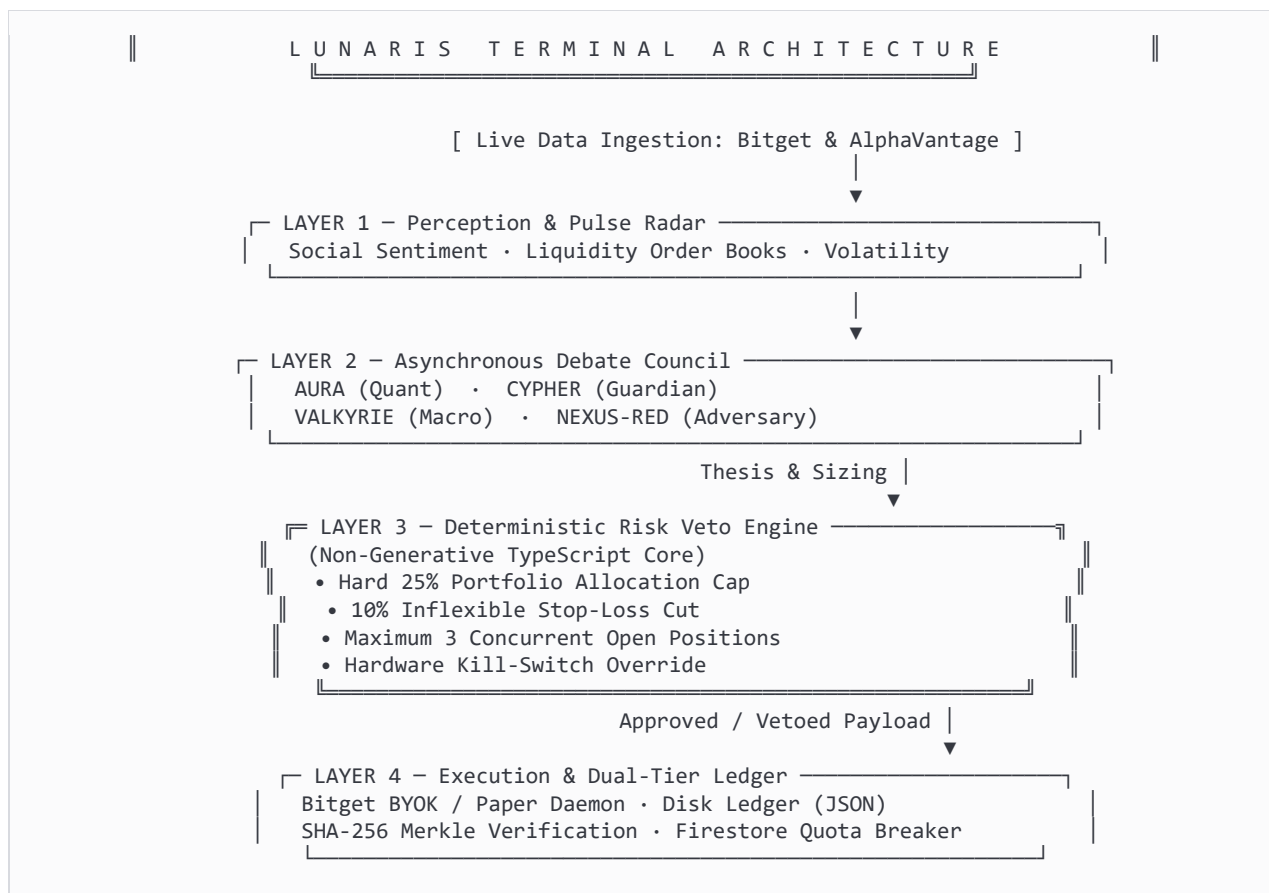
Traditional “AI trading bots” suffer from four critical architectural defects:

- **Sycophancy & Confirmation Bias:** LLMs naturally optimize for helpfulness and narrative consistency. If prompted with bullish signals, they rationalize long positions while ignoring countervailing downside indicators.
- **Hallucination Under High Volatility:** In black-swan scenarios (e.g., flash crashes, de-pegs, exchange outages), LLMs extrapolate fictional price floors or recommend buying into catastrophic drawdowns.
- **Absence of Hard Invariants:** Giving an LLM direct API access to order sizing allows stochastic token generation to dictate capital allocation — a fatal flaw when a prompt injection or parsing error produces an outsized position.
- **Opaque, Unauditable Execution:** Most systems lack verifiable logs, obscuring whether paper-traded returns stem from real-time market data or retroactive in-memory fabrication.

2. Core Architectural Philosophy: The Separation of Intelligence and Authority

The fundamental thesis of LUNARIS is that generative models should advise, but only mathematical invariants may execute.





3. Deep Dive into System Layers

3.1 Layer 1: Multi-Modal Ingestion & Pulse Radar

Before any trading thesis is formed, LUNARIS aggregates multi-source telemetry:

- **Level-2 Depth Order Books:** Real-time bid/ask pressure, depth imbalance, and spread dynamics across Bitget pairs (BTC, ETH, SOL) and equity benchmarks (NVDA, TSLA, AAPL).
- **Social & Volume Pulse Radar:** Ingests social sentiment velocity, institutional order flow patterns, and 24-hour volume changes to evaluate whether price action is driven by organic spot accumulation or speculative derivatives leverage.

3.2 Layer 2: The Four-Persona Adversarial Debate Council

Instead of querying a single prompt, LUNARIS convenes a structured quorum modeled after institutional investment committees. The council consists of four specialized personas:

- **Aura (Lead Quantitative Strategist):** Evaluates technical indicators (EMA crossovers, RSI extremes, Bollinger Band squeezes, and Order Book Volume Imbalance). Seeks asymmetric risk/reward setups.



- **Cypher (Capital Guardian & Risk Officer):** Scrutinizes leverage, portfolio exposure, correlation clustering, and cash drag. Champions capital preservation above all else.
- **Valkyrie (Macro & Geopolitical Analyst):** Integrates central bank liquidity cycles, Treasury yields, cross-asset correlations (e.g., BTC/NVDA coupling), and macro volatility regimes.
- **Nexus-Red (The Red Team Adversary):** The system's primary defense against sycophancy. Its explicit directive is to find flaws in Aura's thesis — constructing worst-case scenarios, liquidity dry-ups, regulatory intervention vectors, and bear traps.

Quorum Consensus Protocol

A trade cannot reach the execution pipeline without a supermajority score from the council. If Nexus-Red surfaces an unhedged liquidation risk or macro catalyst that Cypher validates, the proposal is rejected before it ever reaches the order desk.

COUNCIL DELIBERATION OUTPUT

```
// Architectural Flow: The Council Deliberation Output
interface CouncilQuorumOutcome {
  consensusScore: number; // 0 to 100%
  recommendedAction: 'BUY' | 'SELL' | 'HOLD';
  targetAllocationPct: number; // Proposed by AI (subject to Layer 3 veto)
  dissentingArguments: string[]; // Documented risks surfaced by Nexus-Red
  quorumApproved: boolean;
}
```

3.3 Layer 3: The Deterministic Risk Veto Engine

Even when the AI Council achieves a 100% unanimous consensus, it possesses zero authority to execute directly. The proposal must pass through a strict mathematical filter implemented in pure, non-generative TypeScript (lib/riskVeto.ts):

- **The 25% Allocation Ceiling:** Even if the AI proposes deploying 50% of available capital, the veto engine automatically clamps the trade to 25% max allocation.
- **Non-Negotiable Stop-Loss (10%):** All open positions carry an automated, deterministic stop-loss trigger. The AI is not consulted when a stop-loss is triggered — the execution daemon liquidates the position unconditionally.
- **Maximum Concurrent Positions Limit:** Clamped to a maximum of 3 concurrent open positions to prevent cross-asset correlation contagion.
- **Deterministic Kill Switch:** A persistent emergency circuit breaker capable of instantly freezing autonomous execution, cancelling pending queue entries, and offloading open risk.

3.4 Layer 4: Execution, Dual-Tier Persistence & Cryptographic Merkle Proofs

To guarantee institutional credibility, LUNARIS implements a resilient, multi-tiered persistence and auditing layer:



- **Bitget BYOK (Bring Your Own Keys) Integration:** Users can toggle between verified sandbox/paper trading and live Bitget API execution with localized credential encryption.
- **Dual-Tier Server Ledger:** Trade histories are maintained on persistent server storage (data/audit_trades.json and data/autopilot_state.json), ensuring state survival across server cold-starts and container migrations.
- **Firestore Quota Circuit Breaker:** To insulate against free-tier API exhaustion (e.g., Spark plan write caps), LUNARIS incorporates an intelligent circuit breaker with local storage persistence and automated write-batching, preventing network stalls while continuing real-time audits.

Cryptographic Trade Proofs (SHA-256 Merkle Roots)

Every executed trade generates a deterministic hash calculated over its immutable parameters. Auditors and judges can inspect the input string and recalculate the SHA-256 digest in real time to verify that past returns are mathematically tamper-evident.

4. UI/UX Design System: The Industrial Cyber-Aesthetic

LUNARIS rejects the generic “SaaS template” aesthetic — purple gradients, generic drop shadows, and center-aligned marketing copy — in favor of a specialized mission-control trading console.

- **Typographic Hierarchy:** Monospaced tabular data (JetBrains Mono / Space Grotesk styling) ensures alignment of financial figures, execution deltas, and timestamp micro-seconds.
- **Palette Logic:** High-contrast dark canvas (#08090C, #0C0E14) paired with high-contrast functional color-coding, detailed below.

Signal Color		Semantic Role
#00F0FF Cyan		Quant algorithms, market data streams, active telemetry
#10B981 Emerald		Reconciled profits, quorum BUY consensus, validated signatures
#F43F5E Rose / Crimson		Stop-loss triggers, Nexus-Red adversarial critiques, risk vetoes
#F59E0B Amber		Circuit breakers, quota mitigations, liquidity depth warnings

- **Real-Time Visual Components:** Interactive Liquidity Depth Heatmaps, 3D WebGL Wireframe Visualizers, Dynamic Candlestick Canvas charts with live volume profiles, and a real-time Audit Deck.

5. Comparative Evaluation: Conventional Bot vs. LUNARIS



Dimension	Typical AI Trading Bot ("AI Slop")	LUNARIS Terminal
Cognitive Architecture	Single prompt ("Predict tomorrow's price")	4-Persona Quorum with active Red-Team adversary
Risk Containment	LLM decides order sizing; prone to hallucination	Non-overrideable deterministic risk veto engine
Sycophancy Mitigation	None; model conforms to user prompt bias	Dedicated adversarial persona (Nexus-Red)
Auditability	Ephemeral, opaque, in-memory arrays	SHA-256 Merkle root verification & server ledger
Infrastructure Resilience	Fails on API rate limits or quota drops	Dual-tier fallback with persistent circuit breaker
Exchange Integration	Mock data or unverified API stubs	Production Bitget BYOK sandbox & live endpoints

6. Real-World Applications & Future Horizons

- **Autonomous Hedge Desk Execution:** Operating as an automated scout that detects cross-asset divergences (e.g., tech equity rallies preceding crypto liquidity expansion) and drafts risk-vetted proposals for human review.
- **Auditable Quantitative Paper-Trading:** Providing students, quantitative researchers, and retail traders with a zero-capital simulation environment with cryptographic integrity.
- **On-Chain DAO Treasury Risk Management:** Applying the multi-agent council protocol to on-chain multisig transactions, where adversarial agents debate treasury allocations before funds are deployed.

7. Conclusion

LUNARIS Terminal proves that autonomous AI in quantitative finance does not have to be an unpredictable black box. By pairing adversarial multi-agent intelligence with inflexible deterministic mathematical gates, LUNARIS delivers an autonomous trading architecture that remains disciplined, transparent, and resilient even during severe market stress.

PROJECT LINKS

Live application: <https://lunaristerminal.vercel.app/>

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